

(0) File Naming Explanation:

1. bc1-background

2. f1-foremesh

3. NightCaseTotal-Workbench about the total analysis with the Fluid Flow(Fluent) function

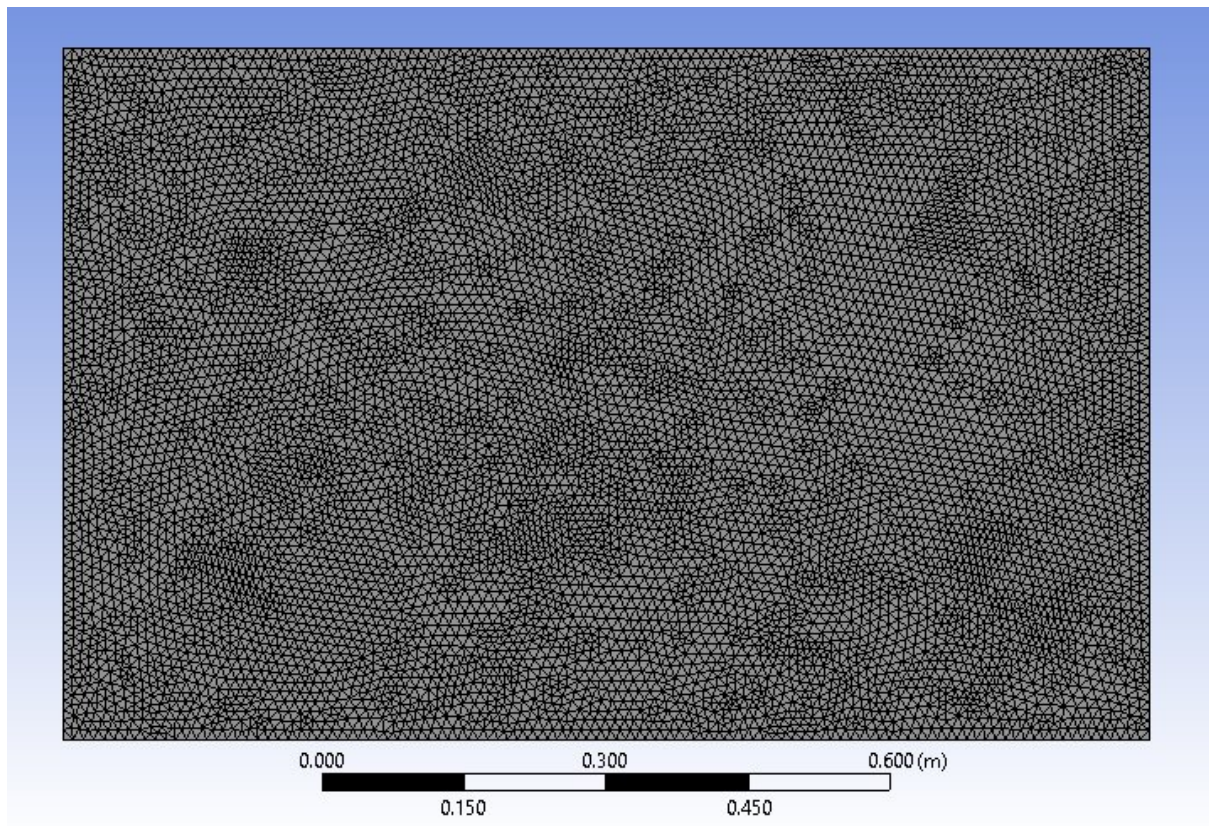
4. vortex.scm-Define the vorticity function to interpret the animated analysis after simulation

5. udf-Define the trajectory of the wing flapping motion

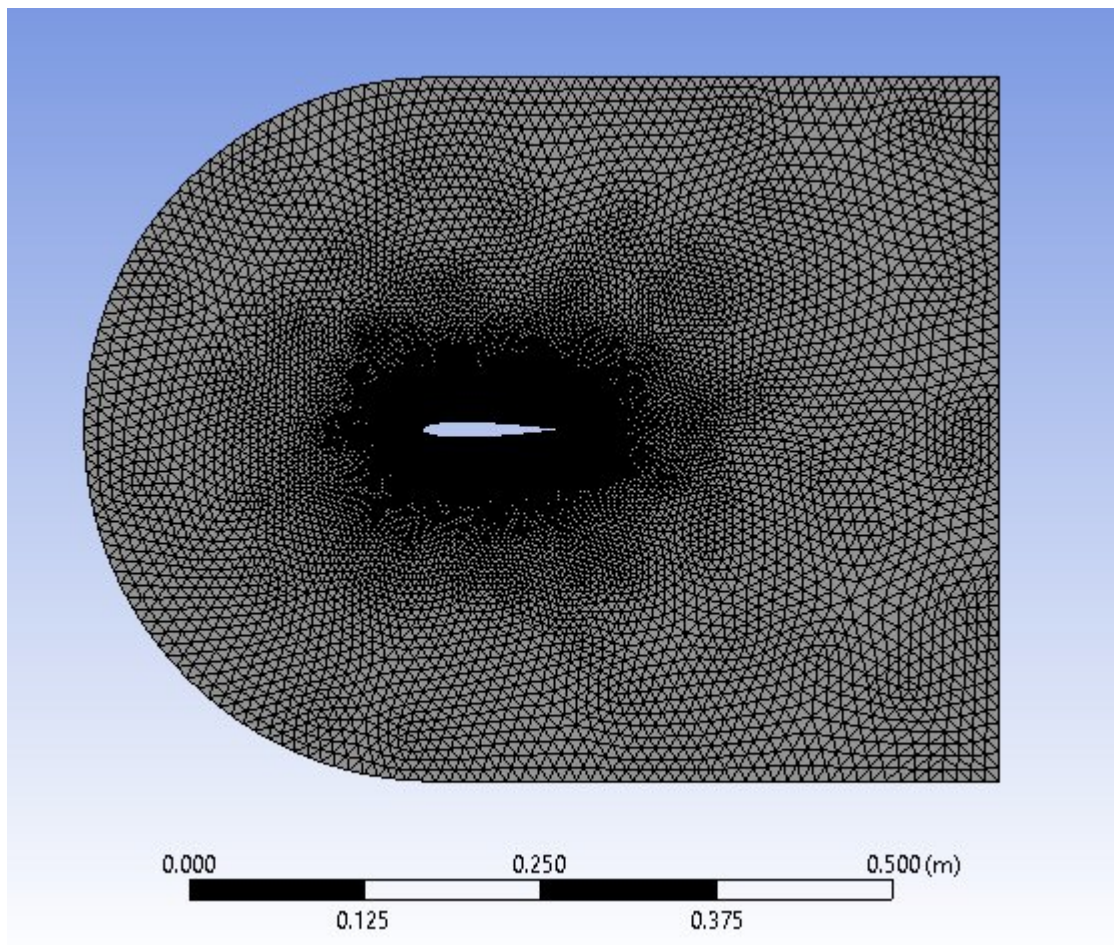
6. 1.cas.h5-the result of the Fluent Analysis

(1) Overset mesh

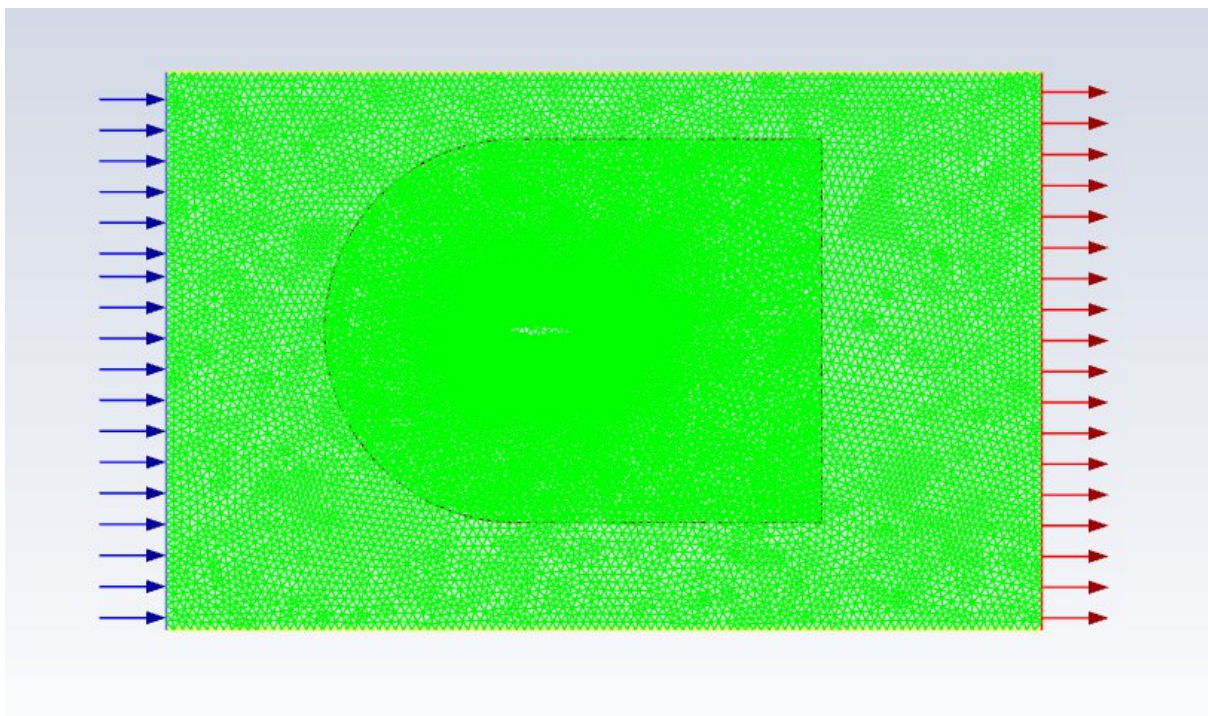
Background mesh:



Foremesh mesh:



Total mesh with background mesh and foremesh mesh:



(2) Animation Video of Simulation Result

Without Vector.GIF

https://github.com/Pika-Zhihao/2.Flapping-Wing-Simulation-Fluid-Structure-Interaction/blob/main/without_vector.gif

With Vector.GIF

https://github.com/Pika-Zhihao/2.Flapping-Wing-Simulation-Fluid-Structure-Interaction/blob/main/with_vector.gif

(3) Data file source and interpretation

WingPointData-naca0012.txt

<https://github.com/Pika-Zhihao/2.Flapping-Wing-Simulation-Fluid-Structure-Interaction/blob/main/naca0012.txt>

Vortex Function Definition-vortex.scn

<https://github.com/Pika-Zhihao/2.Flapping-Wing-Simulation-Fluid-Structure-Interaction/blob/main/vortex.scn>

[This custom field function, ‘custom-function-vorz’, calculates: Two-dimensional vorticity:

$$\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

1. analyze the rotational intensity in a flow field;
2. important in problems such as airfoil flapping, flow around, and vortex generation;
3. used for post-processing, plotting, and animation.]

Definition of wing flapping motion equation-udf.c

<https://github.com/Pika-Zhihao/2.Flapping-Wing-Simulation-Fluid-Structure-Interaction/blob/main/udf.c>

Set the parameters according to the macros in the Fluent official manual and reference 7

Fluent official manual:

https://ansyshelp.ansys.com/public/account/secured?returnurl=/////Views/Secured/corp/v242/en/flu_udf/flu_udf.html

reference 7:

<https://github.com/Pika-Zhihao/2.Flapping-Wing-Simulation-Fluid-Structure-Interaction/blob/main/Reference%207%20Effect%20of%20motion%20trajectory%20on%20the%20aerodynamic%20performance%20of%20a%20flapping%20airfoil.pdf>

[This UDF means:

- 1. Simulating a rigid body (such as an airfoil) performing periodic up-and-down flapping motions in the Y direction;**
- 2. Simultaneously performing periodic rotations in the Z direction;**
- 3. Applicable to scenarios such as biomimetic flapping wings and aircraft flapping simulations.]**

(4) Shortcomings and optimization ideas

1. From the overall simulation operation:

In the simulated animation, the mesh boundary between the foreground and background meshes is not very smooth. Overall, this does not affect the analysis results, but it can still be optimized. Since this is just a simple experiment try, insufficient simulation data has not been used for comparison and verification of mesh independence and other subsequent validations. Further in-depth research in this area is needed for more comprehensive analysis.

2. From the set data parameters:

The data settings in udf.c and vortex.scm are just for experimentation. For a more in-depth analysis of vortex separation during wing flapping motion,

detailed data comparisons are needed, such as comparing vortex distribution under different motion trajectories or the same trajectory at different angular velocities. Additionally, control group experiments can be conducted to analyze the inlet and outlet parameters. Furthermore, the vortex formula has been simplified in its application. To further enhance the reliability of the simulation, more parameters Such as wing mechanical properties could be added for constraint. Analysis of physical quantities such as temperature could also be incorporated. This can be more consistent with the actual experimental conditions.